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Article type : Randomized Clinical Trial

The effects of periodontal treatment on diabetic patients: the DIAPERIO randomized controlled trial

Running title: Periodontitis and diabetes: DIAPERIO RCT

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This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process, which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1111/jcpe.13003

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Keywords: Periodontal Diseases; Periodontal Medicine; Periodontitis; Randomized Controlled Trial; Diabetes Mellitus; Quality of Life.

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Conflict of Interest and Source of Funding: The authors hereby explicitly state that there are no conflicts of interest in connection with this article. This study was supported by a grant from the French Ministry of Health Clinical Research Hospital Program "2008" (Programme Hospitalier de Recherche Clinique inter-régional 0803108 [PHRCI]). The equipment required for scaling and root planing (supersonic motor, periodontal inserts and handpieces) was provided free of cost by the ACTEON® Group (Merignac, France). We would like to thank Oral-B France, whose endowment helped to finance part- time staff for this project.

Abstract

Aim: To assess whether periodontal treatment can lead to clinical, glycemic control and quality of life improvements in metabolically unbalanced diabetic patients (Type 1 or Type 2) diagnosed with periodontitis.

Methods: In this open-labeled randomized controlled trial, diabetic subjects (n=91) were given "immediate" or "delayed" periodontal treatment (full-mouth non-surgical scaling and root planing, systemic antibiotics, and oral health instructions). The main outcome was the effect on glycated hemoglobin (HbA_{1C}) and fructosamine levels. The General Oral Health Assessment Index and the SF-36 index were used to assess quality of life (QoL).

Results: Periodontal health significantly improved after periodontal treatment ($p<0.001$). Periodontal treatment seemed to be safe but had no significant effects on glycemic control based on HbA_{1C} (adjusted mean difference with a 95% confidence interval (aMD) of 0.04[−0.16;0.24]) and fructosamine levels (aMD 5.0[−10.2;20.2]). There was no obvious evidence of improvement in general QoL after periodontal treatment. However, there was significant improvement in oral health-related QoL (aMD 7.0[2.4;11.6], p=0.003).

Conclusion: Although periodontal treatment showed no clinical effect on glycemic control in this trial, important data was provided to support periodontal care among diabetic patients. Periodontal treatment is safe and improves oral health-related QoL in patients living with diabetes. ISRCTN15334496.

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Clinical relevance:

Scientific rationale: Despite intensive research over the past decades, current evidence on the effect of periodontal therapy on glycemic control remains unclear.

Principal findings: Periodontal treatment does not improve glycemic control in Type 1 or Type 2 diabetic patients but significantly improves oral health.

Practical implications: Both periodontitis and diabetes are complex and multifactorial chronic diseases. Therefore, it is likely that person-centered approaches involving both oral health care professionals and diabetologists will contribute to optimal patient care.

Introduction

While many systemic diseases have been considered to be associated with periodontitis, the link between diabetes and periodontitis has been the subject of the most intensive research over the past decades (Monsarrat et al., 2016). Both periodontitis and diabetes are highly prevalent in modern societies (Kassebaum et al., 2017; Murray et al., 2012) and have a negative impact on quality of life (Haag et al., 2017; Zhang and Gregg, 2017).

Current models support the idea of a bi-directional relationship (Taylor et al., 2013). Periodontal therapy might help to reduce local inflammation, and consequently, the putative systemic entry of bacterial by-products (*e.g.* lipopolysaccharides from gram-negative bacterial walls) and pro-inflammatory molecules.

Despite the fact that a reduced HbA_{1C} level still indicates benefits of periodontal treatment on glycemic control in diabetic patients (Madianos and Koromantzos, 2018), effect sizes reported in successive meta-analyses over the past ten years have been globally decreasing (Vergnes, 2015). In a recently updated Cochrane systematic review, it was pointed out that some methodological improvements should be made in future trials (Simpson et al. 2010; Simpson et al. 2015). There is a particular need for specific data concerning Type 1 diabetic subjects and the adjunctive use of antibiotics in periodontal treatment. An improvement could also be made in the assessment of patient-centered outcomes related to quality of life (Darré et al. 2008; Simpson et al. 2015; Madianos and Koromantzos 2018).

The main objective of this randomized trial was to assess whether periodontal treatment with the adjunctive use of amoxicillin could lead to clinical improvements in metabolically unbalanced diabetic patients (Type 1 or Type 2) diagnosed with periodontitis. Secondary objectives were to determine the safety of periodontal treatment and its effect on periodontal health, oral health and general quality of life.

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Materials and Methods

The CONSORT guidelines (Supplementary Table 1) were followed in this study and the full study protocol was published (Vergnes et al., 2009) and registered before the first participants were enrolled (ISRCTN15334496).

Study design

The DIAPERIO (DIAbetes and PERIOdontitis) trial is a prospective, open-labeled, randomized controlled clinical trial. It is a two-center trial with two parallel groups (the treatment group was given "immediate" periodontal treatment versus the control group, which was given "delayed" periodontal treatment, with a balanced 1:1 treatment allocation) and with a follow-up of 3 months.

The trial registered as ID-0803108 was approved by an independent ethics committee (CPP Sud-Ouest Outremer 1 on 23-FEB-2009) and regulatory bodies (AFSSAPS N°2009-A01467-48 on 02-12-2009; CCTIRS on 15-JAN-2009). A 3-month deferral for the periodontal treatment was not considered a loss of opportunity for the participants, since the average waiting time for the treatment of chronic periodontitis is more than 3 months in routine hospital practice. The trial was conducted in compliance with the provisions of the Declaration of Helsinki and French Good Clinical Practice guidelines for periodontal treatment in diabetic patients. All participants gave their written informed consent.

Study participants

Patients were recruited for screening between February 23, 2010 and August 19, 2015 in the Diabetology Departments of two teaching hospitals in southwestern France (Toulouse-Rangueil and Bordeaux-Haut-Lévèque). They were then directed to the respective Dental Health Departments for inclusion (V1).

Subjects were included if diagnosed with one of the two main types of diabetes mellitus (Type 1 or Type 2) at least one year before the date of inclusion, were uncontrolled on the date of inclusion (HbA_{1C} level between 7.0% and 9.5%), and if their diabetes treatment had remained unchanged over the previous 3 months (no change of dosage, formulation or method of administration). To be included, patients were required to have at least six permanent natural teeth and a diagnosis of periodontitis attested by the presence of at least 4 teeth with at least one probed site with a periodontal pocket depth (PPD) ≥ 4 mm and a clinical attachment level (CAL) ≥ 3 mm.

Clinical investigators collected sociodemographic data, medical histories, and information related to medical treatments, lifestyle and oral hygiene habits.

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Study intervention

Periodontal treatment included non-surgical scaling and root planing (SRP), systemic antibiotics (amoxicillin 2 g/day for 7 days after periodontal treatment) and providing a supply of oral hygiene products and oral health instructions. The antibiotic therapy was prescribed to comply with French recommendations for the prevention of local infection following invasive oral treatments in patients with uncontrolled diabetes (FSOS, 2012). Furthermore, systemic antibiotics in nonsurgical periodontal therapy for diabetic subjects could lead to a modest additional improvement in periodontal parameters (Grellmann et al. 2016; Souto et al. 2018). SRP was performed over the 10-day period following the inclusion visit for the immediate treatment group (Supplementary Figure 1). SRP was performed in a single 2-hour session. Dental surgeons who were specially trained by an experienced periodontist prior to the launch of the study performed full-mouth ultrasonic and hand instrument debridement. Local anesthesia was administered if necessary. SRP was accompanied by subgingival irrigation with an antiseptic mouthwash (chlorhexidine 0.12%). After SRP, subjects received personalized oral health instructions (OHI) and a free pack containing all the dental care products needed for 3 months of periodontal maintenance.

Control intervention

Delayed periodontal treatment for the control group was performed with exactly the same procedures at the end of the 3-month follow-up period (Supplementary Figure 1) for ethical reasons. Therefore, the control group was a true “no treatment” control arm for which no periodontal procedure was performed during the follow-up.

Study outcomes

Venous blood samples for biological markers and laboratory tests were collected and immediately sent to the respective Clinical Biochemistry Departments. Samples were analyzed in a blind.

Primary outcome: the primary outcome of the DIAPERIO trial was the change in HbA_{1C} between the inclusion visit (V1) and the 3-month follow-up visit (V4 endpoint). HbA_{1C} levels (normal range 4.0 - 6.0%) were measured by high-performance liquid chromatography (G7HPLC Tosoh Analyzer, Tosoh Bioscience, CA, USA).

Secondary outcomes: we also considered the change in fructosamine levels between V1 and V4 and between V1 and the intermediate visit in 3 weeks (V3) (Supplementary Figure 1). Fructosamine (normal range 165 - 285 µmol/L) was measured with a Cobas 8000 analyzer (Roche Diagnostics, France).

A constant pressure probe (Florida Probe®; Gainesville, FL, USA) was used for full-mouth periodontal charting of 6 measurements per tooth (Gupta et al., 2015). This included CAL, PPD and bleeding on probing (BoP). Investigators were trained in periodontal assessment (CAL, PPD and BoP) before the launch of the study with the overall inter-rater kappa reaching 0.9. With the individual periodontal measurements, a global periodontal inflammatory burden was also estimated at the participant

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level. This took into account the severity, extent and activity of periodontitis. The total surface area of the gingival epithelium (Periodontal Epithelial Surface Area – PESA) and the total surface area of inflamed gingiva (Periodontal Inflammation Surface Area – PISA) were calculated (Nesse et al., 2009). We also classified periodontitis at baseline according to the extent and the severity (Armitage, 1999). Extent was characterized as localized (≤ 30% of sites involved) or generalized (> 30%). Severity was characterized based on CAL, and categorized as mild (mean: 1–2 mm of periodontal attachment loss), moderate (mean: 3–4 mm of periodontal attachment loss) or severe (mean: ≥ 5 mm of periodontal attachment loss). No X-rays were taken for the trial.

Patient-centered outcomes consisted of quality of life assessments. This included both a general quality of life (health-related quality of life - HRQoL) assessment with the MOS 36-Item Short Form Health Survey (SF-36), and an oral health-related quality of life (OHRQoL) assessment with the General Oral Health Assessment Index (GOHAI). Both are self-administered questionnaires that have been validated in French (Tubert-Jeannin et al., 2003; Perneger et al., 1995). GOHAI was analyzed as an additive score (Add-GOHAI) and according to its 3 dimensions (Blaizot et al., 2013), and the SF-36 according to its 8 dimensions of health (Walters, 2009; Ware and Sherbourne, 1992).

All adverse health events during the 3-month follow-up period were recorded for both groups after each participant was interviewed at the beginning of each follow-up visit. Special emphasis was placed on reporting any potential adverse oral health events (e.g. dental hypersensitivity, tooth staining or acute pain).

The methods used to analyze plasma cytokine concentrations (IL-1β, IFN-γ, IL-10, IL-6, MCP-1, TNF-α, RANTES) can be found in the Supplementary Methods (supplementary research called "DIAPERIO-Bio", conducted in a subgroup of DIAPERIO patients).

Statistical analyses

Sample size calculation was based on the detection of a 0.5% change from baseline HbA_{1C} between the two groups. Based on the assumption of a standard deviation of 1.0% with a two-sided test at a 5% significance level, 64 participants per group would yield a power of 80%. The initial target sample size was 150 participants (75 per center, with independent randomization between the two centers) with an anticipated dropout rate of 15%. Details of the randomization process are provided in the open-access protocol (Vergnes et al., 2009).

The analysis was carried out on the intention-to-treat (ITT) population, *i.e.* all randomized subjects, regardless of whether they were present at the intermediate visits or complied with treatment (antibiotic therapy or oral hygiene). The data were also analyzed using the *per protocol* (PP) approach, which restricted the analysis to participants who complied with the treatment protocol (present at all visits and brushed teeth at least twice a day during the follow-up for the treatment group).

The main analysis was a comparison between the two arms regarding changes in HbA_{1C} and fructosamine levels during follow-up. The mean differences were estimated with a 95% confidence interval and were analyzed by covariance analysis (ANCOVA), adjusted for the baseline HbA_{1C} level and the clinical center (aMD). Residual values were tested for an approximated normal distribution.

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In addition, five binary outcomes were created: a relative decrease ≥ 10% in HbA_{1C} level, a three-month level of < 7%, an absolute decrease ≥ 0.5%, a relative increase ≥ 10% and an absolute increase ≥ 0.5%. These variables were used in logistic regression models adjusted for the baseline HbA_{1C} level and the clinical center.

Statistical analyses were performed at two levels. First, Type 1 and Type 2 diabetes were analyzed together. The rationale for this combination was that there is no evidence available concerning a differential biological effect of periodontal treatment on glycemic control between the two types of diabetes. In fact, both types are associated with the same increasing risk of common periodontal issues (Kaur et al. 2015), with slight differences in the inflammatory mechanisms of diabetes mellitus-associated periodontitis (Aspriello et al. 2011), but no evidence of different biologic hypotheses regarding the effect of periodontal treatment on glycemic control (Polak and Shapira 2018). Secondly, analyses were performed according to the type of diabetes (because they should also be considered as two different entities (Madianos and Koromantzos 2018)).

To assess the efficacy of periodontal treatment for periodontal health and other secondary outcomes, variables were analyzed with a model similar to that specified for the primary outcome. The proportion of subjects in both groups who had one or more adverse events was compared using Fisher's exact test.

Results

Participants

A total of 180 individuals were screened and 91 were randomized between February 2010 and August 2015 (Figure 1, Table 1). In September 2015, when the number needed to treat had been reached in the Toulouse center (n=82), the scientific committee of the trial decided to end recruitment as this was considered futile (Jitlal et al., 2012). This was based on an interim analysis which showed that the conditional power was low (threshold below 35%), and that the effect size was very close to a null effect (mean difference close to zero, Table 2). Due to organizational issues, the Bordeaux center recruited only 9 patients, but data from these participants were nevertheless included in the analysis.

The control and treatment groups consisted of 46 (35 Type 1 and 11 Type 2) and 45 subjects (32 Type 1 and 13 Type 2), respectively. Two patients in the control group and one patient in the treatment group were lost during follow-up. Therefore, the ITT analysis was carried out for 44 patients in each group for the HbA_{1C} outcome (Figure 1, Table 2). For the PP analysis, 42 subjects from the control group (31 Type 1 and 11 Type 2) and 34 subjects from the treatment group (25 Type 1 and 9 Type 2) were considered (Supplementary Table 2).

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Glycemic control outcomes

No statistically significant differences in HbA_{1C} levels were found between the groups (Table 2, Figure 2a), neither for Type 1 and Type 2 diabetes combined, nor according to diabetes type. The ANCOVA, adjusted on the baseline value and on the clinical center, also showed no difference between the two groups (aMD 0.04 [−0.16; 0.24], p = 0.67) for Type 1 and Type 2 diabetes together. No difference was detected for fructosamine levels at the intermediate visit or in 3 months, after adjustment on the baseline value and on the clinical center (aMD −8.0 [−22.0; 5.9], p = 0.26 and 5.0 [−10.2; 20.2], p = 0.51, respectively). No difference was found in the PP analysis (Supplementary Table 3). The computed odds ratio for the binary HbA_{1C} outcomes failed to demonstrate a difference between the control and treatment groups (Supplementary Table 4). The DIAPERIO-Bio supplemental analysis detected no evidence of any effect of the periodontal treatment on biological markers (Supplementary Table 5).

Periodontal outcomes

Mean PPD, mean CAL and % BoP sites improved significantly 3 months after SRP for Type 1 and Type 2 diabetes together (an adjusted mean difference with a 95% confidence interval (aMD) with −0.32 mm [−0.43; −0.21], −0.27 mm [−0.44; −0.09] and −13.6% [−19.2; −8.0], respectively. The mean PISA was also reduced after periodontal treatment (aMD −170.7 mm^2 [−231.3; −110.1]) (Table 3, Figure 2b). No significant difference was noted in periodontal improvement according to diabetes duration, diabetes treatment, smoking status, age or BMI ($p > 0.05$). No difference in periodontal improvement was detected between the two types of diabetes or according to baseline HbA_{1C} (p = 0.99, p = 0.58, respectively). Similar results were obtained for the PP analysis (Supplementary Table 3).

Quality of life and adverse effect outcomes

The mean sum of GOHAI scores (Add-GOHAI) increased in the treatment group compared to the control group (aMD 7.0 [2.4;11.6]) due to improvements in the “psychological impacts” and “pain and discomfort” domains (Table 4). Similar results were obtained for the PP analysis (Supplementary Table 3). A significant effect was noted in the General Health domain of general quality of life (SF-36) for Type 1 diabetes (aMD 6.3 [0.4; 12.2], p = 0.04).

During the 3-month follow-up period, 23 subjects in the control group and 22 subjects in the treatment group reported having a health problem that might affect the course of the clinical trial ($p > 0.99$). 15 and 18 control and treatment subjects respectively experienced oral disorders (p = 0.37). The treatment group experienced more dental hypersensitivity (p = 0.03) but with a tendency towards less diffuse pain (p = 0.07). There were no other differences in adverse health experiences between the two groups.

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Discussion

In this 3-month randomized controlled clinical trial, periodontal treatment had no effect on glycemic control. HbA_{1C} and fructosamine levels were assessed together and separately in uncontrolled Type 1 and Type 2 diabetic patients, with no effect noted. However, significant positive effects of periodontal treatment on oral health-related quality of life were observed.

Our results should be discussed in the context of previous randomized controlled trials with a similar design (*i.e.* the effect on HbA_{1C} in 3-4 months as the main outcome, adjunctive use of antibiotics in the treatment group and a control group with no active treatment). The results of this trial are consistent with current evidence that the effect of periodontal treatment on HbA_{1C} in 3-4 months is, at best, clinically modest. With regards to the adjunctive use of amoxicillin, our results are consistent with the paradoxical evidence in the literature. In fact, adjunctive use of antibiotics in the periodontal treatment of diabetic patients seems to be effective in improving the clinical and microbiological outcomes of SRP (Miranda et al. 2014). However, there is no consistent evidence that the addition of antimicrobials (whether antiseptics, antibacterials or antibiotics) to SRP is more beneficial than SRP alone in terms of HbA_{1C} levels (Simpson et al. 2015).

Regardless of the various factors that might explain the heterogeneity of results between studies, there is a clear trend of decreasing effect sizes according to the meta-analyses. One example is the results from the 2010 version of the Cochrane systematic review compared to those from the 2015 version (Simpson et al. 2010; Simpson et al. 2015). Some recent trials with "negative" results explain this trend (Gay et al., 2014; Mizuno et al., 2017), in particular the large-scale American DPTT trial (Engebretson et al., 2013). This trend of decreasing effect sizes over time might be surprising, but it is a well-known phenomenon in clinical research. Ioannidis described the "Proteus phenomenon" (causing quite a stir in the scientific community) according to which extreme research claims rapidly alternate with extreme opposite refutations (Ioannidis and Trikalinos, 2005). For the subject at hand, the time lag in publishing negative findings might also explain the decrease in effect sizes. It is also possible that well-designed studies take longer to complete, especially because of the difficult and time-consuming task of arranging standardized follow-up appointments with outpatient diabetic participants. To start with, the inclusion period for our clinical trial, for example, was 5 years because the randomization of 91 subjects required screening of 180 patients in 2-hour pre-inclusion visits, which included a complete periodontal check-up (required for standardized recruitment). In addition, the enrolled subjects were seen at least 4 times, which means that over 500 appointments were required to complete the study.

In short, factors that might explain the heterogeneity of results obtained by different trials are countless: the type of interventions (in both groups), the type of diabetes considered, the severity of periodontitis, HbA_{1C} levels at the time of enrollment, the inclusion of smokers, socio-economical levels and compliance with the treatment regimen are just a few examples. This makes it very difficult to methodically synthesize current knowledge in order to design future research, and even more so in order to establish clear clinical recommendations. In our opinion, in this context where there is a vast amount of complex information, new and complementary approaches should be used. In fact, traditional systematic reviews are limited by their restriction to specific comparisons that do not reflect the variety of practices and therapeutic options. In the first instance, Knowledge

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Discovery in Databases (KDD), already used in the field of periodontal medicine (Monsarrat et al. 2016), would enable the application of novel methods and algorithms for the mapping, analysis and interpretation of complex biomedical data, aimed at identifying testable hypotheses and building realistic models (Holzinger and Jurisica 2014). KDD applied to databases such as PubMed could provide new insights in the diabetes - periodontal diseases relationship. Secondly, there is a need for improvement in data sharing practices among periodontology researchers. Publishers play an important role in encouraging and mandating the availability of data, to allow meaningful re-analysis and re-use (Naudet et al. 2018). Through such a practice, individual patient-level data from existing databases could be combined. Thirdly, at the clinical level, person-centered approaches are useful in decision-making resulting in a balance of complex scientific evidence with complex patient perspectives (Apelian et al. 2017).

This brings us back to the present study, which has some strengths and limitations regarding the methods, sample, analyses, and transferability. However, an important and original aspect of our trial is the establishment of an optimal periodontal treatment (comprised of SRP, OHI, antiseptics, and antibiotics), versus no treatment, to increase the probability of determining any effect of periodontal treatment on glycemic control. This maximization is unique in the field since previous studies either provided sub-optimal periodontal treatments (e.g. SRP and OHI alone, such as in Gay et al. 2014) or potentially effective periodontal procedures in control groups (e.g. OHI, such as in the study by Engebretson *et al.* 2013). In view of the fact that periodontal inflammation greatly diminished after periodontal treatment in our study, futility cannot be explained by an ineffective intervention. Neither can our "negative" result be explained by any periodontal procedure in the control group, since our control group was a true "no treatment" control arm. One drawback of our "optimal periodontal treatment" approach is that recent research shows that it is the combination of amoxicillin and metronidazole that yields the best clinical results in nonsurgical periodontal therapy for diabetic subjects (Souto et al. 2018).

A further strength of our study is that to our knowledge, DIAPERIO is one of the very few randomized controlled trials that assess the effect of periodontal treatment on a substantial sample of Type 1 diabetic patients. By providing data on both types of diabetes, our trial tends to support previous studies that indicate that the relationship between diabetes mellitus and periodontal disease is not totally dependent on the type of diabetes (Demmer et al. 2012; Kaur et al. 2015).

The DIAPERIO trial is also the first to investigate the effect of periodontal treatment on oral health-related quality of life in diabetic patients. The GOHAI was used to evaluate the OHRQoL (Tubert-Jeannin et al., 2003).This brings the patient's daily experience to the forefront (Blaizot et al., 2013), and provides the possibility to determine functional problems, discomfort and pain (Nikbin et al., 2014). In fact, several studies have established a link between diabetes and low OHRQoL (Sadeghi et al., 2014; Nikbin et al., 2014; Azogui-Lévy et al., 2017). Periodontitis is known to affect OHRQoL by increasing the severity and extent of periodontal disease (Buset et al., 2016). Conversely, non-surgical periodontal treatment can improve immediate and long-term OHRQoL (Goel and Baral, 2017), particularly in terms of function, psychology and pain (Shanbhag et al., 2012). We noted significant improvement in the GOHAI score after periodontal treatment, especially for the pain perception and psychological domains, and there was also a tendency towards improvement for the physical functioning domain. However, this finding should be carefully weighed, considering the fact that participants from the intervention group experienced more hypersensitivity symptoms, which is

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a well-known consequence of SRP (Engebretson et al., 2013). Fortunately this side-effect was transient. At the same time, in terms of patients' overall quality of life, diabetes is known to have negative consequences (Al Hayek et al., 2014; Dogan et al., 2016), particularly for those with poor glycemic control (Dogan et al., 2016). Although, there was a positive trend for the "General Health" (vitality, energy, fatigue) and "Physical Functioning" (measurement of limitations during physical activity) components, we found no significant changes in any of the eight domains of the SF-36. This could be explained by the lack of effect of periodontal treatment on glycemic control and biological markers.

Several limitations should also be taken into account. The most important one might be that the DIAPERIO trial was initially designed to include both types of diabetic patients without a pre-defined stratification strategy by diabetes type. The reason was the lack of evidence of any distinctive biological effect of periodontal treatment on glycemic control between the two types of diabetes. Given the low number of participants with Type 2 diabetes and the persisting possibility of different pathogeneses between the two types, we must therefore be cautious when generalizing combined results. However, in analyses where Type 1 and Type 2 diabetes were combined, we found no effect modification by type of diabetes. This suggests that the absence of a relationship between periodontal treatment and diabetes might be unrelated to the type of diabetes. Nonetheless, it is important to note that this interaction analysis might lack power, especially as a result of the low number of Type 2 diabetes participants. In our study the majority of patients had Type 1 diabetes (about 70% of the subjects included, because participants who agreed to enroll in the trial were mainly patients participating in the Hospital diabetes education program, which was not anticipated).

Another limitation is that it is likely that the DIAPERIO participants are not representative of uncontrolled diabetic patients, even within France. Again, enrolment in a hospital setting, with mainly hospital-based diabetes education programs, favors the inclusion of Type 1 diabetic patients from quite a high socio-economic level. Considering that there is a strong socio-economic component to periodontitis in both diabetic and non-diabetic French populations (Azogui-Lévy and Rochereau, 2014), it is possible that our inclusion process resulted in patients with less severe and less active periodontitis being selected. The DIAPERIO participants had an initial mean PPD of approximately 2.3 mm and 30% BoP, which suggests better periodontal health compared to the 3.3 mm and 60% BoP in DPTT participants, for example. Despite this selection of participants from a mid to high socio-economic level, and the fact that dental consultations are fully reimbursed in France, it was surprising to note that participants reported much fewer visits to the dentist than to other medical professions such as the ophthalmologist or podiatrist. In addition, the ratio between the two centers was unbalanced, which was problematic in terms of the external validity of the trial, although our results were adjusted for the center effect afterwards.

Finally, there are two further limitations in this study. The first is that we did not identify oral hygiene indicators during periodontal examinations. While plaque indicators are important to assess participant compliance with oral health hygiene instructions, we did not consider this data to be essential because the focus was on the ITT analysis. The second is that the SF-36 questionnaire might not be the best method to measure the effect of treatment on the QoL of diabetes patients. A recent randomized trial reported a positive effect of periodontal treatment on QoL in Type 2 diabetes patients, assessed with the Diabetes Therapy-Related QoL (Mizuno et al., 2017). This questionnaire

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assesses the influence of diabetes treatment on patient QoL, regardless of the treatment method. Unfortunately, it was developed in 2012 when our trial had already started (Ishii, 2012).

Conclusion

Diabetes and periodontitis are both complex and multifactorial diseases. As such, demonstrating the therapeutic effect of one on the other through clinical trials is not simple. Although we cannot affirm that periodontal treatment has no effect on glycemic control, regardless of the type of diabetes considered, it is likely that this effect is modest at best. In any event, periodontal treatment reduces oral inflammation, is safe, improves the oral health-related QoL of diabetic patients, and might have an effect on overall QoL. Therefore, referral to a dentist is essential and should be an integral part of diabetes management (Sadeghi et al., 2014).

The DIAPERIO Study Group

Members of the DIAPERIO (DIAbetes and PERIOdontitis) study group:

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Acknowledgments

We are very grateful to L. Darré and O. Sechoy-Balussou for their help during the design of the trial, NE. Ghelamallah, V. Portet, D. Tardivo for their help during subject recruitment, the entire Bordeaux CHU pre-analytical team and the Toulouse biochemistry department for managing the blood samples, F. Foulong and the other members of the Toulouse pre-analytical team for the management of the DIAPERIO-bio serum bank, the genotoul anexplo phenotyping platform (UMS006 - ANEXPLO/CREFRE), the entire Rangueil Diabetology Department team of nurses for their warm welcome and assistance in recruiting patients, the entire Rangueil Dentistry Department team of nurses for subject sampling (M. Barollo, M. Capdrot, M. Dulherm, P. Fourcade, G. Joly, B. Galaup, C. Sergues, J. Sudres, V. Pizzuto, I. Vetel, M. Mariotto), Prof V. Dupuis for her support in the

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implementation of the trial at the Bordeaux Teaching Hospital Dentistry Department, Karen Anderson-Dutin (Amplus Traduction) for proofreading and, of course, all the trial participants.

References

Al Hayek AA, Robert AA, Al Saeed A, Alzaid AA, Al Sabaan FS. 2014. Factors Associated with Health-Related Quality of Life among Saudi Patients with Type 2 Diabetes Mellitus: A Cross-Sectional Survey. Diabetes Metab J. 38:220–229. doi:10.4093/dmj.2014.38.3.220.

Apelian N, Vergnes JN, Hovey R, Bedos C. 2017. How can we provide person-centred dental care? Br Dent J. 223:419–424. doi:10.1038/sj.bdj.2017.806.

Armitage GC. 1999. Development of a classification system for periodontal diseases and conditions. Ann Periodontol Am Acad Periodontol. 4:1–6. doi:10.1902/annals.1999.4.1.1.

Aspriello SD, Zizzi A, Tirabassi G, Buldreghini E, Biscotti T, Faloia E, Stramazzotti D, Boscaro M, Piemontese M. 2011. Diabetes mellitus-associated periodontitis: differences between type 1 and type 2 diabetes mellitus. J Periodontal Res. 46:164–169. doi:10.1111/j.1600-0765.2010.01324.x.

Azogui-Lévy S, Dray-Spira R, Attal S, Hartemann A, Anagnostou F, Azerad J. 2017 Nov 15. Factors associated to oral health related quality of life in patients with diabetes. Aust Dent J. doi:10.1111/adj.12577.

Azogui-Lévy S, Rochereau T. 2014. [Dental health and dental care according diabetic status; results from 2008 ESPS study]. Rev Epidemiol Sante Publique. 62:329–337. doi:10.1016/j.respe.2014.09.005.

Blaizot A, Monsarrat P, Constantin A, Vergnes J-N, de Grado GF, Nabet C, Cantagrel A, Sixou M. 2013. Oral health-related quality of life among outpatients with rheumatoid arthritis. Int Dent J. 63:145–153. doi:10.1111/idj.12023.

This article is protected by copyright. All rights reserved.

Accepted Article

Buset SL, Walter C, Friedmann A, Weiger R, Borgnakke WS, Zitzmann NU. 2016. Are periodontal diseases really silent? A systematic review of their effect on quality of life. J Clin Periodontol. 43:333–344. doi:10.1111/jcpe.12517.

Darré L, Vergnes J-N, Gourdy P, Sixou M. 2008. Efficacy of periodontal treatment on glycaemic control in diabetic patients: A meta-analysis of interventional studies. Diabetes Metab. 34:497–506. doi:10.1016/j.diabet.2008.03.006.

Demmer RT, Holtfreter B, Desvarieux M, Jacobs DR, Kerner W, Nauck M, Völzke H, Kocher T. 2012. The influence of type 1 and type 2 diabetes on periodontal disease progression: prospective results from the Study of Health in Pomerania (SHIP). Diabetes Care. 35:2036–2042. doi:10.2337/dc11-2453.

Dogan H, Harman E, Kocoglu H, Sargin G. 2016. Can metabolic control variables of diabetic patients predict their quality of life? J Am Soc Hypertens JASH. 10:81–88. doi:10.1016/j.jash.2015.11.014.

Engebretson SP, Hyman LG, Michalowicz BS, Schoenfeld ER, Gelato MC, Hou W, Seaquist ER, Reddy MS, Lewis CE, Oates TW, et al. 2013. The effect of nonsurgical periodontal therapy on hemoglobin A1c levels in persons with type 2 diabetes and chronic periodontitis: a randomized clinical trial. JAMA. 310:2523–2532. doi:10.1001/jama.2013.282431.

FSOS. 2012. Management of mouth foci of infection. Recommandations of the French society of Oral Surgery.

Gay IC, Tran DT, Cavender AC, Weltman R, Chang J, Luckenbach E, Tribble GD. 2014. The effect of periodontal therapy on glycaemic control in a Hispanic population with type 2 diabetes: a randomized controlled trial. J Clin Periodontol. 41:673–680. doi:10.1111/jcpe.12268.

Accepted Article

This article is protected by copyright. All rights reserved.

Goel K, Baral D. 2017. A Comparison of Impact of Chronic Periodontal Diseases and Nonsurgical Periodontal Therapy on Oral Health-Related Quality of Life. Int J Dent. 2017:9352562. doi:10.1155/2017/9352562.

Grellmann AP, Sfreddo CS, Maier J, Lenzi TL, Zanatta FB. 2016. Systemic antimicrobials adjuvant to periodontal therapy in diabetic subjects: a meta-analysis. J Clin Periodontol. 43:250–260. doi:10.1111/jcpe.12514.

Gupta N, Rath SK, Lohra P. 2015. Comparative evaluation of accuracy of periodontal probing depth and attachment levels using a Florida probe versus traditional probes. Med J Armed Forces India. 71:352–358. doi:10.1016/j.mjafi.2012.02.018.

Haag DG, Peres KG, Balasubramanian M, Brennan DS. 2017. Oral Conditions and Health-Related Quality of Life: A Systematic Review. J Dent Res. 96:864–874. doi:10.1177/0022034517709737.

Holzinger A, Jurisica I. 2014. Knowledge Discovery and Data Mining in Biomedical Informatics: The Future Is in Integrative, Interactive Machine Learning Solutions. In: Interactive Knowledge Discovery and Data Mining in Biomedical Informatics. Springer, Berlin, Heidelberg. (Lecture Notes in Computer Science). p. 1–18. [accessed 2018 May 8]. https://link.springer.com/chapter/10.1007/978-3-662-43968-5_1.

Ioannidis JPA, Trikalinos TA. 2005. Early extreme contradictory estimates may appear in published research: the Proteus phenomenon in molecular genetics research and randomized trials. J Clin Epidemiol. 58:543–549. doi:10.1016/j.jclinepi.2004.10.019.

Ishii H. 2012. Development and psychometric validation of the Diabetes Therapy-Related QOL (DTR-QOL) questionnaire. J Med Econ. 15:556–563. doi:10.3111/13696998.2012.665111.

Jitlal M, Khan I, Lee SM, Hackshaw A. 2012. Stopping clinical trials early for futility: retrospective analysis of several randomised clinical studies. Br J Cancer. 107:910–917. doi:10.1038/bjc.2012.344.

This article is protected by copyright. All rights reserved.

Accepted Article

Kassebaum NJ, Smith AGC, Bernabé E, Fleming TD, Reynolds AE, Vos T, Murray CJL, Marcenes W, Abyu GY, Alsharif U, et al. 2017. Global, Regional, and National Prevalence, Incidence, and Disability-Adjusted Life Years for Oral Conditions for 195 Countries, 1990–2015: A Systematic Analysis for the Global Burden of Diseases, Injuries, and Risk Factors. J Dent Res. 96:380–387. doi:10.1177/0022034517693566.

Kaur PK, Narula SC, Rajput R, K Sharma R, Tewari S. 2015. Periodontal and glycemic effects of nonsurgical periodontal therapy in patients with type 2 diabetes stratified by baseline HbA1c. J Oral Sci. 57:201–211. doi:10.2334/josnusd.57.201.

Madianos PN, Koromantzos PA. 2018. An update of the evidence on the potential impact of periodontal therapy on diabetes outcomes. J Clin Periodontol. 45:188–195. doi:10.1111/jcpe.12836.

Miranda TS, Feres M, Perez-Chaparro PJ, Faveri M, Figueiredo LC, Tamashiro NS, Bastos MF, Duarte PM. 2014. Metronidazole and amoxicillin as adjuncts to scaling and root planing for the treatment of type 2 diabetic subjects with periodontitis: 1-year outcomes of a randomized placebo-controlled clinical trial. J Clin Periodontol. 41:890–899. doi:10.1111/jcpe.12282.

Mizuno H, Ekuni D, Maruyama T, Kataoka K, Yoneda T, Fukuhara D, Sugiura Y, Tomofuji T, Wada J, Morita M. 2017. The effects of non-surgical periodontal treatment on glycemic control, oxidative stress balance and quality of life in patients with type 2 diabetes: A randomized clinical trial. PloS One. 12:e0188171. doi:10.1371/journal.pone.0188171.

Monsarrat P, Blaizot A, Kémoun P, Ravaud P, Nabet C, Sixou M, Vergnes J-N. 2016. Clinical research activity in periodontal medicine: a systematic mapping of trial registers. J Clin Periodontol. 43:390-400. doi:10.1111/jcpe.12534.

Murray CJL, Vos T, Lozano R, Naghavi M, Flaxman AD, Michaud C, Ezzati M, Shibuya K, Salomon JA, Abdalla S, et al. 2012. Disability-adjusted life years (DALYs) for 291 diseases and injuries in 21

Accepted Article

This article is protected by copyright. All rights reserved.

regions, 1990-2010: a systematic analysis for the Global Burden of Disease Study 2010. Lancet Lond Engl. 380:2197–2223. doi:10.1016/S0140-6736(12)61689-4.

Naudet F, Sakarovitch C, Janiaud P, Cristea I, Fanelli D, Moher D, Ioannidis JPA. 2018. Data sharing and reanalysis of randomized controlled trials in leading biomedical journals with a full data sharing policy: survey of studies published in The BMJ and PLOS Medicine. BMJ. 360:k400.

Nesse W, Linde A, Abbas F, Spijkervet FKL, Dijkstra PU, de Brabander EC, Gerstenbluth I, Vissink A. 2009. Dose-response relationship between periodontal inflamed surface area and HbA1c in type 2 diabetics. J Clin Periodontol. 36:295–300. doi:10.1111/j.1600-051X.2009.01377.x.

Nikbin A, Bayani M, Jenabian N, Khafri S, Motallebnejad M. 2014. Oral health-related quality of life in diabetic patients: comparison of the Persian version of Geriatric Oral Health Assessment Index and Oral Health Impact Profile: A descriptive-analytic study. J Diabetes Metab Disord. 13:32. doi:10.1186/2251-6581-13-32.

Perneger TV, Leplège A, Etter JF, Rougemont A. 1995. Validation of a French-language version of the MOS 36-Item Short Form Health Survey (SF-36) in young healthy adults. J Clin Epidemiol. 48:1051–1060.

Polak D, Shapira L. 2018. An update on the evidence for pathogenic mechanisms that may link periodontitis and diabetes. J Clin Periodontol. 45:150–166. doi:10.1111/jcpe.12803.

Sadeghi R, Taleghani F, Farhadi S. 2014. Oral health related quality of life in diabetic patients. J Dent Res Dent Clin Dent Prospects. 8:230–234. doi:10.5681/joddd.2014.41.

Shanbhag S, Dahiya M, Croucher R. 2012. The impact of periodontal therapy on oral health-related quality of life in adults: a systematic review. J Clin Periodontol. 39:725–735. doi:10.1111/j.1600-051X.2012.01910.x.

Accepted Article

This article is protected by copyright. All rights reserved.

Simpson TC, Needleman I, Wild SH, Moles DR, Mills EJ. 2010. Treatment of periodontal disease for glycaemic control in people with diabetes. Cochrane Database Syst Rev.:CD004714. doi:10.1002/14651858.CD004714.pub2.

Simpson TC, Weldon JC, Worthington HV, Needleman I, Wild SH, Moles DR, Stevenson B, Furness S, Iheozor-Ejiofor Z. 2015. Treatment of periodontal disease for glycaemic control in people with diabetes mellitus. Cochrane Database Syst Rev. 11:CD004714. doi:10.1002/14651858.CD004714.pub3.

Souto MLS, Rovai ES, Ganhito JA, Holzhausen M, Chambrone L, Pannuti CM. 2018. Efficacy of systemic antibiotics in nonsurgical periodontal therapy for diabetic subjects: a systematic review and meta-analysis. Int Dent J. doi:10.1111/idj.12384.

Taylor JJ, Preshaw PM, Lalla E. 2013. A review of the evidence for pathogenic mechanisms that may link periodontitis and diabetes. J Clin Periodontol. 40 Suppl 14:S113-134. doi:10.1111/jcpe.12059.

Tubert-Jeannin S, Riordan PJ, Morel-Papernot A, Porcheray S, Saby-Collet S. 2003. Validation of an oral health quality of life index (GOHAI) in France. Community Dent Oral Epidemiol. 31:275–284.

Vergnes J-N. 2015. Review finds periodontal treatment has short term benefits for diabetics. Evid Based Dent. 16:78–79. doi:10.1038/sj.ebd.6401114.

Vergnes J-N, Arrivé E, Gourdy P, Hanaire H, Rigalleau V, Gin H, Sédarat C, Dorignac G, Bou C, Sixou M, et al. 2009. Periodontal treatment to improve glycaemic control in diabetic patients: study protocol of the randomized, controlled DIAPERIO trial. Trials. 10:65. doi:10.1186/1745-6215-10-65.

Walters SJ. 2009. Randomized Controlled Trials. In: Quality of Life Outcomes in Clinical Trials and Health-Care Evaluation. John Wiley & Sons, Ltd. p. 181–215. [accessed 2016 Jun 10]. http://onlinelibrary.wiley.com/doi/10.1002/9780470840481.ch8/summary.

Accepted Article

This article is protected by copyright. All rights reserved.

Ware JE, Sherbourne CD. 1992. The MOS 36-item short-form health survey (SF-36). I. Conceptual framework and item selection. Med Care. 30:473–483.

Zhang P, Gregg E. 2017. Global economic burden of diabetes and its implications. Lancet Diabetes Endocrinol. 5:404–405. doi:10.1016/S2213-8587(17)30100-6.

Accepted Article

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Tables

Table 1: Sociodemographic and medical characteristics for the intention to treat subjects at inclusion

	Type 1 diabetes				*Type 2 diabetes*			
Sociodemographic characteristics	**N**	**Control group**	**N**	**Treatment group**	**N**	**Control group**	**N**	**Treatment group**
Age (years)	35	53.7 ± 13.8	32	50.9 ± 9.4	11	63.1 ± 4.0	13	68.3 ± 9.3
Women	35	15 (43%)	32	13 (41%)	11	2 (18%)	13	5 (38%)
Level of education								
Elementary school	32	0 (0%)	32	0 (0%)	11	1 (9%)	13	2 (16%)
Middle school		9 (28%)		11 (35%)		1 (9%)		5 (38%)
High school		10 (31%)		5 (15%)		4 (36%)		1 (8%)
University level		13 (41%)		16 (50%)		5 (46%)		5 (38%)
Professional activities								
Farmers, operators: primary sector	33	1 (3%)	32	0 (0%)	11	0 (0%)	13	0 (0%)
Artisans, traders and entrepreneurs		0 (0%)		4 (12%)		0 (0%)		0 (0%)
Executives, intellectual professions		7 (22%)		3 (9%)		0 (0%)		0 (0%)
Intermediate occupations		3 (9%)		5 (16%)		0 (0%)		0 (0%)
Blue-collar employees, workers		10 (30%)		8 (25%)		1 (9%)		1 (8%)
Retirees		6 (18%)		5 (16%)		6 (55%)		9 (69%)
Other people without work		6 (18%)		7 (22%)		4 (36%)		3 (23%)

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Native country								
France	33	31 (94%)	32	29 (91%)	11	7 (64%)	13	9 (69%)
Other		2 (6%)		3 (9%)		4 (36%)		4 (31%)

Medical characteristics	**N**	**Control group**	**N**	**Treatment group**	**N**	**Control group**	**N**	**Treatment group**
Body mass index (kg/m²)	35	25.5 ± 4.3	31	24.9 ± 4.1	11	31.4 ± 4.4	13	27.4 ± 3.5
Perceived stress	33	14 (42%)	32	12 (38%)	11	3 (27%)	13	5 (38%)
Smokers	33	8 (24%)	31	7 (23%)	11	0 (0%)	13	2 (15%)
Smoking history (years)	6	24.2 ± 8.0	7	22.9 ± 8.7	0	-	2	35.0 ± 7.1
Number of cigarettes a day	7	15.0 ± 3.8	8	12.5 ± 9.5	0	-	2	3.0 ± 1.4
Drug treatments (except diabetes)[a]								
Digestive system and metabolism	35	2 (6%)	32	4 (13%)	11	2 (18%)	13	1 (8%)
Blood and blood-forming organs		6 (18%)		4 (13%)		6 (55%)		3 (23%)
Cardiovascular system		19 (58%)		16 (50%)		9 (82%)		7 (30%)
Urogenital system		1 (3%)		2 (6%)		0 (0%)		1 (8%)
Systemic hormones		6 (18%)		7 (22%)		2 (18%)		3 (23%)
Systemic anti-infective		0 (0%)		2 (6%)		0 (0%)		0 (0%)
Antineoplastic		1 (3%)		0 (0%)		0 (0%)		0 (0%)
Musculoskeletal system		1 (3%)		0 (0%)		1 (9%)		2 (16%)
Nervous system		3 (9%)		4 (13%)		0 (0%)		4 (32%)
Respiratory tract system		0 (0%)		1 (3%)		0 (0%)		1 (8%)
Sensory organs		0 (0%)		1 (3%)		0 (0%)		0 (0%)
No treatment		13 (37%)		8 (25%)		1 (9%)		2 (16%)
Diabetes history (years)	33	25.2 ± 11.0	32	25.0 ± 13.9	11	19.9 ± 13.0	13	18.1 ± 11.2
HbA_{1C} (%)	35	7.83 ± 0.64	32	7.84 ± 0.65	11	7.78 ± 0.52	13	7.96 ± 0.84
Last modification of treatment (months)	33	70.6 ± 49.4	30	79.5 ± 85.5	10	35.2 ± 48.2	12	55.8 ± 52.2
Last ophthalmology visit (months)	33	9.3 ± 17.4	32	10.6 ± 21.9	11	8.1 ± 8.6	12	21.1 ± 37.6
Last podiatry visit (months)	27	9.1 ± 18.4	25	20.0 ± 30.9	7	9.3 ± 14.6	9	10.0 ± 12.0

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Type of diabetes treatment								
Oral antidiabetic drugs		0 (0%)		1 (3%)		6 (55%)		4 (31%)
Insulin	33	28 (85%)	32	28 (88%)	11	0 (0%)	13	3 (23%)
Insulin and oral antidiabetic drugs		5 (15%)		3 (9%)		5 (45%)		6 (46%)

Dental characteristics	**N**	**Control group**	**N**	**Treatment group**	**N**	**Control group**	**N**	**Treatment group**
Last visit to the dentist (months)	33	14.3 ± 15.0	31	10.6 ± 11.2	10	25.9 ± 44.0	12	39.1 ± 43.9
Brushing frequency								
At least twice a day		21 (64%)		24 (75%)		8 (73%)		7 (54%)
Once a day	33	12 (36%)	32	7 (22%)	11	3 (27%)	13	5 (38%)
Sporadically		0 (0%)		1 (3%)		0 (0%)		1 (8%)
Interproximal hygiene frequency								
At least once a day		6 (18%)		2 (6%)		1 (9%)		1 (8%)
Sporadically	33	4 (12%)	32	7 (22%)	11	6 (55%)	13	3 (23%)
Never		23 (70%)		23 (72%)		4 (36%)		9 (69%)
Mouthwash use frequency								
At least twice a day		2 (6%)		2 (6%)		0 (0%)		0 (0%)
Once a day		2 (6%)		4 (13%)		1 (9%)		3 (23%)
Sporadically	33	15 (46%)	32	17 (53%)	11	9 (82%)	13	8 (62%)
Never		14 (42%)		9 (28%)		1 (9%)		2 (15%)

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Frequency of visits to the dentist								
At least twice a year		10 (30%)		5 (16%)		3 (27%)		3 (23%)
Once a year		12 (37%)		19 (59%)		5 (46%)		4 (31%)
Once every two years	33	6 (18%)	32	5 (16%)	11	1 (9%)	13	2 (15%)
Less than once every two years		5 (15%)		3 (9%)		2 (18%)		4 (31%)
Previously informed about the importance of oral health in diabetic patients	33	9 (27%)	32	10 (31%)	11	1 (9%)	13	0 (0%)
Periodontitis extent and severity								
Generalized - severe		2 (6%)		6 (19%)		3 (27%)		3 (24%)
Generalized - moderate	35	12 (34%)	32	10 (31%)	11	5 (46%)	13	5 (38%)
Localized - severe		14 (40%)		12 (38%)		3 (27%)		5 (38%)
Localized - moderate		7 (20%)		4 (12%)		0 (0%)		0 (0%)
Number of missing teeth	35	7.1 ± 4.4	35	6.7 ± 3.2	11	11.2 ± 4.4	35	11.2 ± 4.6

Results are presented as the mean ± the standard deviation for quantitative outcomes and as the frequency (%) for qualitative outcomes.
[a] Drug treatments were classified using the Anatomical Therapeutic Chemical (ATC) classification.

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Table 2: Effects of the intervention on the biological markers related to diabetes for the intention to treat subjects

Medical outcomes	N	Control group Mean ± SD	N	Treatment group Mean ± SD	p [a]	adjusted Mean Difference [b] aMD [CI95%]	p [c]
HbA_{1C} (%) at V4							
Type 1 diabetes	33	7.82 ± 0.64	32	7.78 ± 0.65	0.72	−0.02 [−0.24; 0.20]	0.87
Type 2 diabetes	11	7.65 ± 0.72	12	8.07 ± 0.90	0.28	0.24 [−0.27; 0.75]	0.34
All types of diabetes	44	7.77 ± 0.66	44	7.86 ± 0.73	0.78	0.04 [−0.16; 0.24]	0.67
Fructosamine (µmol/l) at V3							
Type 1 diabetes	30	325.2 ± 50.5	27	327.7 ± 51.6	0.90	−11.4 [−28.8; 6.0]	0.19
Type 2 diabetes	9	299.3 ± 42.3	10	324.4 ± 88.6	0.51	−0.34 [−22.0; 21.3]	0.97
All types of diabetes	39	319.2 ± 49.4	37	326.8 ± 62.3	0.67	−8.0 [−22.0; 5.9]	0.26
Fructosamine (µmol/l) at V4							
Type 1 diabetes	30	320.4 ± 40.1	28	321.4 ± 49.4	0.88	0.41 [−14.4; 15.2]	0.96
Type 2 diabetes	11	291.6 ± 39.9	9	346.7 ± 148.5	0.46	−0.32 [−34.3; 33.7]	0.98
All types of diabetes	41	312.7 ± 41.6	37	327.5 ± 82.8	0.73	5.0 [−10.2; 20.2]	0.51
Weight (kg) at V4							
Type 1 diabetes	30	77.7 ± 19.5	26	72.5 ± 13.8	0.42	−0.15 [−1.4; 1.1]	0.80
Type 2 diabetes	10	94.4 ± 16.0	12	77.9 ± 14.7	0.02	0.52 [−2.2; 3.2]	0.69
All types of diabetes	40	81.9 ± 19.9	38	74.2 ± 14.1	0.11	0.14 [−1.0; 1.3]	0.81

Results are presented as the mean ± the standard deviation.
[a] p-value of the non-parametric Mann-Whitney test for a difference between control and treatment groups.
[b] adjusted mean difference of the V4 or V3 outcome by ANCOVA analysis, adjusted on baseline value (V1) and the clinical center (Bordeaux or Toulouse). A negative value indicates that the value of the treatment group is lower than the control group.
[c] p-value of the adjusted mean difference.

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Table 3: Effects of the intervention on the periodontal outcomes for the intention to treat subjects

Periodontal outcomes	N	Control group Mean ± SD	N	Treatment group Mean ± SD	p [a]	adjusted Mean Difference [b] aMD [CI95%]	p [c]
Periodontal Pocket Depth - PPD (mm) at V4							
Type 1 diabetes	30	2.06 ± 0.37	27	1.90 ± 0.33	0.12	−0.26 [−0.38; −0.14]	<0.001
Type 2 diabetes	10	2.52 ± 0.74	12	1.87 ± 0.56	0.02	−0.38 [−0.65; -0.11]	0.009
All types of diabetes	40	2.18 ± 0.52	39	1.89 ± 0.41	0.009	−0.32 [−0.43; −0.21]	<0.001
Clinical Attachment Loss - CAL (mm) at V4							
Type 1 diabetes	30	2.41 ± 0.65	27	2.38 ± 0.67	0.96	−0.27 [−0.46; −0.08]	0.007
Type 2 diabetes	10	3.19 ± 1.07	12	2.71 ± 1.16	0.20	−0.12 [−0.51; 0.26]	0.51
All types of diabetes	40	2.61 ± 0.83	39	2.48 ± 0.85	0.47	−0.27 [−0.44; −0.09]	0.003
Recession - REC (mm) at V4							
Type 1 diabetes	30	0.35 ± 0.42	27	0.47 ± 0.45	0.25	−0.02 [−0.14; 0.11]	0.80
Type 2 diabetes	10	0.65 ± 0.58	12	0.85 ± 0.93	0.87	0.28 [-0.11; 0.67]	0.15
All types of diabetes	40	0.42 ± 0.46	39	0.59 ± 0.65	0.35	0.06 [−0.08; 0.19]	0.41
Bleeding on Probing - BoP (%) at V4							
Type 1 diabetes	30	26.3 ± 17.9	27	13.5 ± 12.8	0.005	−13.5 [−19.6; −7.4]	<0.001
Type 2 diabetes	10	23.4 ± 14.8	12	14.7 ± 12.3	0.12	−10.3 [-23.5; 2.8]	0.12
All types of diabetes	40	25.6 ± 17.1	39	13.8 ± 12.5	<0.001	−13.6 [−19.2; −8.0]	<0.001
Periodontal Epithelial Surface Area - PESA (mm²) at V4							
Type 1 diabetes	30	937.1 ± 215.0	27	861.1 ± 141.9	0.97	−125.7 [−189.1; −62.2]	<0.001
Type 2 diabetes	10	851.7 ± 242.8	12	680.1 ± 176.8	0.11	−75.5 [−239.4; 88.3]	0.35
All types of diabetes	40	915.8 ± 222.2	39	805.4 ± 173.2	0.03	−113.2 [−171.9; −54.5]	<0.001
Periodontal Inflamed Surface Area - PISA (mm²) at V4							
Type 1 diabetes	30	279.3 ± 214.4	27	130.1 ± 130.8	0.003	−182.0 [−249.3; −114.7]	<0.001
Type 2 diabetes	10	251.2 ± 157.7	12	120.8 ± 116.7	0.04	−139.6 [−267.1; −12.2]	0.03
All types of diabetes	40	272.3 ± 200.2	39	127.3 ± 125.2	<0.001	−170.7 [−231.3; −110.1]	<0.001

Results are presented as the mean ± the standard deviation.
[a] p-value of the non-parametric Mann-Whitney test for a difference between control and treatment groups.
[b] adjusted mean difference of the V4 outcome by ANCOVA analysis, adjusted on baseline value (V1) and the clinical center (Bordeaux or Toulouse). A negative value indicates that the value of the treatment group is lower than the control group.
[c] p-value of the adjusted mean difference.

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Table 4: Effects of the intervention on the quality of life outcomes for the intention to treat subjects

Quality of life outcomes	N	Control group Mean ± SD	N	Treatment group Mean ± SD	p [a]	adjusted Mean Difference [b] aMD [CI95%]	p [c]
GOHAI at V4							
Add-GOHAI (%)							
Type 1 diabetes	30	78.1 ± 13.9	27	88.3 ± 11.1	0.003	10.7 [6.1; 15.3]	<0.001
Type 2 diabetes	10	76.2 ± 11.2	12	70.3 ± 13.8	0.41	−3.3 [-15.9; 9.3]	0.59
All types of diabetes	40	77.7 ± 13.1	39	82.7 ± 14.5	0.07	7.0 [2.4; 11.6]	0.003
Physical functioning (%)							
Type 1 diabetes	31	85.7 ± 14.1	27	94.3 ± 12.2	0.002	6.3 [0.01; 12.7]	0.05
Type 2 diabetes	10	81.4 ± 13.0	12	72.9 ± 13.0	0.19	−2.7 [−16.9; 11.4]	0.69
All types of diabetes	41	84.7 ± 13.8	39	87.7 ± 16.6	0.10	3.4 [−2.3; 9.2]	0.24
Pain and discomfort (%)							
Type 1 diabetes	31	73.3 ± 16.0	27	84.9 ± 17.0	0.006	12.8 [5.2; 20.4]	0.001
Type 2 diabetes	10	63.3 ± 25.4	12	65.4 ± 18.7	1.00	−4.9 [−22.0; 12.2]	0.55
All types of diabetes	41	70.9 ± 18.9	39	78.9 ± 19.6	0.05	8.2 [0.8; 15.6]	0.03
Psychosocial impacts (%)							
Type 1 diabetes	30	74.8 ± 19.8	27	85.4 ± 16.2	0.03	13.2 [5.8; 20.7]	<0.001
Type 2 diabetes	10	80.0 ± 13.3	12	71.3 ± 19.8	0.35	−1.2 [−16.3; 13.9]	0.87
All types of diabetes	40	76.1 ± 18.4	39	81.0 ± 18.4	0.16	9.2 [2.7; 15.7]	0.006
SF-36 at V4							
Physical Functioning (%)							
Type 1 diabetes	30	72.7 ± 16.0	27	79.8 ± 8.4	0.15	4.1 [−1.1; 9.3]	0.12
Type 2 diabetes	11	58.2 ± 19.0	11	57.3 ± 23.5	0.95	5.9 [−5.0; 16.7]	0.27
All types of diabetes	41	68.8 ± 17.8	38	73.3 ± 17.5	0.24	3.4 [−1.4; 8.2]	0.17
Physical Role (%)							
Type 1 diabetes	30	70.8 ± 39.4	27	91.7 ± 25.0	0.04	5.3 [−9.3; 19.9]	0.47
Type 2 diabetes	10	70.0 ± 40.5	12	54.2 ± 45.0	0.76	11.5 [−8.2; 32.2]	0.24
All types of diabetes	40	70.6 ± 39.2	39	80.1 ± 36.4	0.10	3.0 [−8.5; 14.5]	0.60
Bodily Pain (%)							

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Type 1 diabetes	31	77.9 ± 24.9	27	88.4 ± 20.5	0.06	1.8 [−9.5; 13.2]	0.75
Type 2 diabetes	11	55.8 ± 27.3	12	58.6 ± 31.8	0.58	−1.7 [−23.5; 20.2]	0.87
All types of diabetes	42	72.1 ± 27.0	39	79.3 ± 27.9	0.07	−0.5 [−10.6; 9.7]	0.93
General Health (%)							
Type 1 diabetes	31	57.3 ± 14.8	27	63.1 ± 14.4	0.19	6.3 [0.4; 12.2]	0.04
Type 2 diabetes	11	50.0 ± 18.2	11	54.1 ± 17.9	0.62	−3.1 [−13.2; 7.0]	0.53
All types of diabetes	42	55.4 ± 15.8	38	60.5 ± 15.8	0.15	4.1 [−0.9; 9.1]	0.11
Vitality (%)							
Type 1 diabetes	31	61.3 ± 19.2	27	65.6 ± 16.2	0.36	3.5 [−4.0; 11.0]	0.35
Type 2 diabetes	10	59.5 ± 15.4	11	47.7 ± 21.8	0.18	−1.2 [−14.1; 11.8]	0.85
All types of diabetes	41	60.9 ± 18.2	38	60.4 ± 19.5	0.93	0.9 [−5.3; 7.2]	0.77
Social Functioning (%)							
Type 1 diabetes	31	65.6 ± 40.7	27	90.8 ± 16.4	0.07	7.0 [−2.2; 16.2]	0.13
Type 2 diabetes	11	79.6 ± 18.8	12	61.7 ± 32.7	0.30	5.3 [−17.5; 28.0]	0.63
All types of diabetes	42	81.0 ± 20.4	39	81.8 ± 26.1	0.34	−0.8 [−9.7; 8.1]	0.85
Emotional Role (%)							
Type 1 diabetes	31	65.6 ± 40.7	27	90.1 ± 29.0	0.24	−3.5 [−18.6; 11.6]	0.65
Type 2 diabetes	11	81.8 ± 34.6	12	69.5 ± 33.2	0.15	−11.6 [−46.9; 23.6]	0.50
All types of diabetes	42	85.7 ± 30.5	39	83.8 ± 31.4	0.67	−3.1 [−16.3; 10.2]	0.64
Mental Health (%)							
Type 1 diabetes	31	71.0 ± 15.1	27	76.0 ± 17.2	0.24	3.5 [−3.2; 10.2]	0.30
Type 2 diabetes	11	75.3 ± 13.1	11	65.8 ± 23.6	0.51	0.28 [−11.3; 11.8]	0.96
All types of diabetes	42	72.1 ± 14.6	38	73.1 ± 19.5	0.57	0.2 [−5.7; 6.0]	0.96

Results are presented as the mean ± the standard deviation.
[a] p-value of the non-parametric Mann-Whitney test for a difference between control and treatment groups.
[b] adjusted mean difference of the V4 outcome by ANCOVA analysis, adjusted on baseline value (V1) and the clinical center (Bordeaux or Toulouse). A negative value indicates that the value of the treatment group is lower than the control group.
[c] p-value of the adjusted mean difference.

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Figures

Figure 1

Flow diagram of the study, according to the CONSORT statements

Figure 2

(a) Scatter plot of the relationship between HbA_{1C} at V1 and V4 for the control (clear symbols) and treatment groups (black symbols). The linear regression line is shown (solid for the treatment and dashed for the control group). Type 1 and Type 2 diabetes are indicated by circles/a blue line and a triangle/red line, respectively.

(b) Scatter plot of the relationship between PISA at V1 and V4 for the control (clear symbols) and treatment groups (black symbols). The linear regression line is shown (solid for the treatment and dashed for the control group). Type 1 and Type 2 diabetes are indicated by circles/a blue line and a triangle/red line, respectively.

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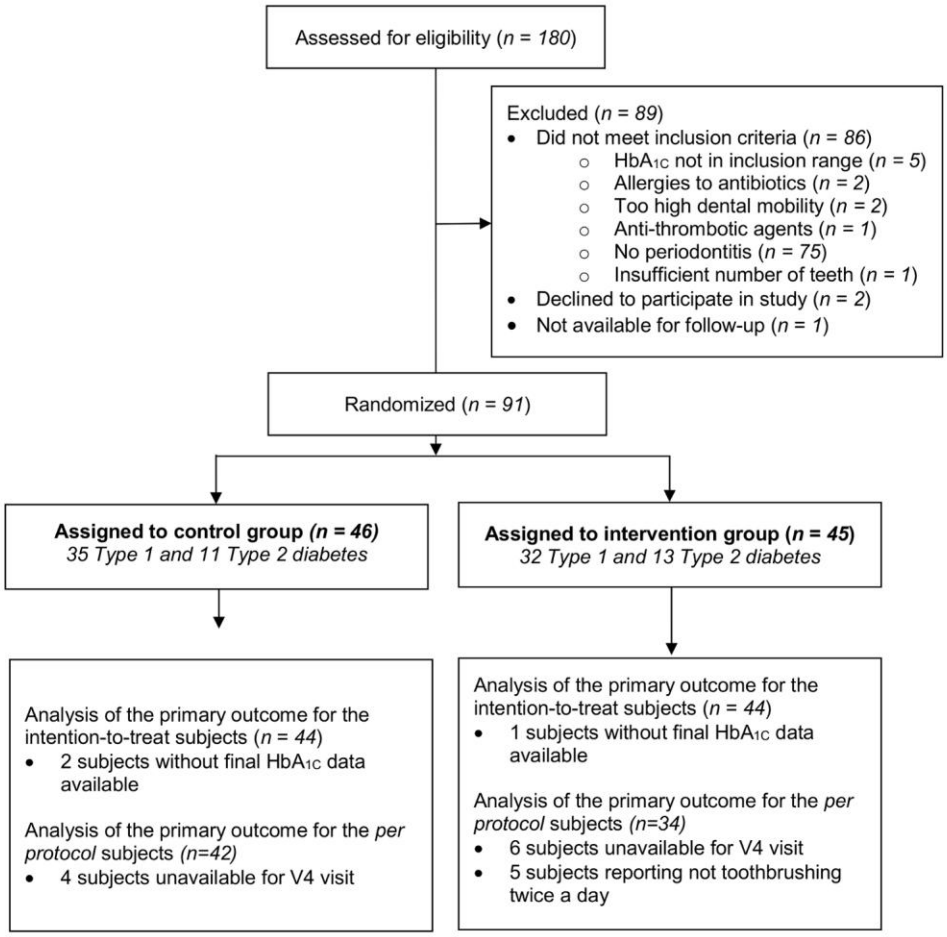

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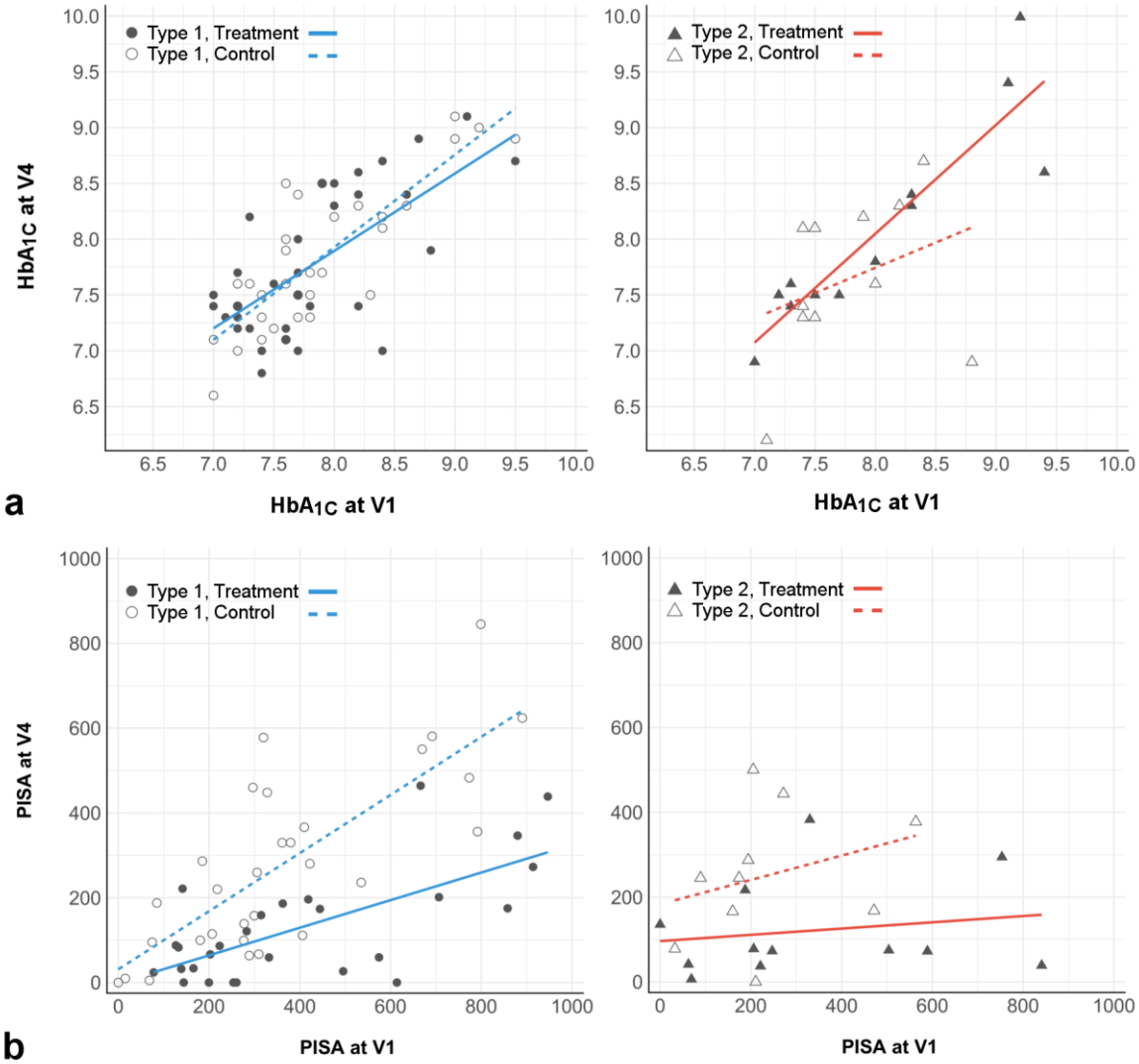

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